

BASIC SQL QUERIES AND AGGREGATE FUNCTIONS

ASSIGNMENT 8

DATABASE MANAGEMENT SYSTEM

ALLIANCE UNIVERSITY

Department of Computer Applications

SUMMITED BY: TINAKAR C

REGISTER NO: 2511021240003

COURSE: BCA

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INTRODUCTION

SQL queries retrieve, filter, sort, and analyze data stored in a database. WHERE, ORDER BY, GROUP BY, and HAVING organize records, while COUNT(), SUM(), AVG(), MIN(), and MAX() calculate values. The following commands are applied to the Online Shopping Management System using MySQL.

1. FILTERING RECORDS USING WHERE

The WHERE clause retrieves records that satisfy a specified condition.

```
SELECT product_id, product_name, price
FROM PRODUCT
WHERE price > 800;
```

Output:

product_id	product_name	price
302	Bluetooth Headphones	1499.00
304	Kitchen Mixer	2499.00
306	Laptop Backpack	1199.00
307	LED Desk Lamp	899.00

This query displays products whose prices are greater than 800.

2. SORTING RECORDS USING ORDER BY

ORDER BY sorts records in ascending (ASC) or descending (DESC) order.

```
SELECT product_name, price
FROM PRODUCT
ORDER BY price DESC;
```

Output:

product_name	price
Kitchen Mixer	2499.00
Bluetooth Headphones	1499.00
Laptop Backpack	1199.00
LED Desk Lamp	899.00
Wireless Mouse	799.00
DBMS Fundamentals	650.00
Cotton T-Shirt	499.00
USB-C Cable	299.00

The most expensive product appears first.

3. GROUPING RECORDS USING GROUP BY

GROUP BY groups records with the same value in a column and is often used with aggregate functions.

```
SELECT order_status, COUNT(*) AS total_orders
FROM ORDERS
GROUP BY order_status;
```

Output:

order_status	total_orders
Delivered	3
Processing	1
Confirmed	1

The query counts orders in each order status.

4. FILTERING GROUPED RESULTS USING HAVING

HAVING filters groups created by GROUP BY.

```
SELECT category_id, COUNT(*) AS total_products
FROM PRODUCT
GROUP BY category_id
HAVING COUNT(*) >= 2;
```

Output:

category_id	total_products
10	3
20	2
30	2

The query displays categories that contain at least two products.

5. AGGREGATE FUNCTIONS

Aggregate functions perform calculations on a set of records.

A. COUNT()

COUNT() counts records.

```
SELECT COUNT(*) AS Total_Customers
FROM CUSTOMER;
```

Output:

Total_Customers
5

There are five customer records.

B. SUM()

SUM() calculates the total of numeric values.

```
SELECT SUM(amount) AS Total_Payment_Amount
FROM PAYMENT;
```

Output:

Total_Payment_Amount
8793.00

This total includes successful and pending payments.

C. AVG()

AVG() calculates the average of numeric values.

```
SELECT AVG(price) AS Average_Product_Price
FROM PRODUCT;
```

Output:

Average_Product_Price
1042.875

This calculates the average price across the eight products.

D. MIN()

MIN() returns the smallest value.

```
SELECT MIN(price) AS Minimum_Price
FROM PRODUCT;
```

Output:

Minimum_Price
299.00

The lowest listed product price is 299.00.

E. MAX()

MAX() returns the largest value.

```
SELECT MAX(price) AS Maximum_Price
FROM PRODUCT;
```

Output:

Maximum_Price
2499.00

The highest listed product price is 2499.00.

CONCLUSION

WHERE filters records, ORDER BY sorts them, GROUP BY creates groups, and HAVING filters grouped results. COUNT, SUM, AVG, MIN, and MAX provide useful summaries for online shopping data.